POWER-TORSION SUBMODULES OF AN INJECTIVE MODULE OVER A NOETHERIAN RING ARE INJECTIVE

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The criterion that a submodule of an injective module is injective exactly when it has no nontrivial essential extension inside it. Invoked at the start of the counted proof (source line 563). Reduces injectivity of the power-torsion submodule to the absence of essential extensions.

Source reference. Original lemma lemma-essential-extensions-in-injective: sources/stacks-dualizing/pinned-original-source.tex, source lines 444–444. Navigation pointer only: the counted text refers to the Stacks item lemma-essential-extensions-in-injective, which is an existing labeled result of this same chapter and is not part of this card. The pointer carries the target's own label as its number and its definition line as its span; its content is not reproduced here.

Lemma 0.1. *Let R be a Noetherian ring. Let I be an injective R-module.*

(1) *Let $f \in R$. Then $E = \bigcup I[f^n] = I[f^\infty]$ is an injective submodule of I.*
(2) *Let $J \subset R$ be an ideal. Then the J-power torsion submodule $I[J^\infty]$ is an injective submodule of I.*

Proof. We will use Lemma lemma-essential-extensions-in-injective to prove (1). Suppose that $E \subset E' \subset I$ and that E' is an essential extension of E. We will show that $E' = E$. If not, then we can find $x \in E'$ and $x \notin E$. Let $J = \{a \in R \mid ax \in E\}$. Since R is Noetherian, we may write $J = (g_1, \ldots, g_t)$ for some $g_i \in R$. By definition E is the set of elements of I annihilated by powers of f, so we may choose integers n_i so that $f^{n_i} g_i x = 0$. Set $n = \max\{n_i\}$. Then $x' = f^n x$ is an element of E' not in E and is annihilated by J. Set $J' = \{a \in R \mid ax' \in E\}$ so $J \subset J'$. Conversely, we have $a \in J'$ if and only if $ax' \in E$ if and only if $f^m a x' = 0$ for some $m \geq 0$. But then $f^m a x' = f^{m+n} a x$ implies $ax \in E$, i.e., $a \in J$. Hence $J = J'$. Thus

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$J = J' = \text{Ann}(x')$, so $Rx' \cap E = 0$. Hence E' is not an essential extension of E, a contradiction.

To prove (2) write $J = (f_1, \ldots, f_t)$. Then $I[J^\infty]$ is equal to

$$(\ldots((I[f_1^\infty])[f_2^\infty])\ldots)[f_t^\infty]$$

and the result follows from (1) and induction. $\square$